

Program-8

Implement hierarchal clustering .

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster

# Generate sample data
X, _ = make_blobs(n_samples=20, centers=3, random_state=42)

# Perform hierarchical clustering
linked = linkage(X, method='ward')

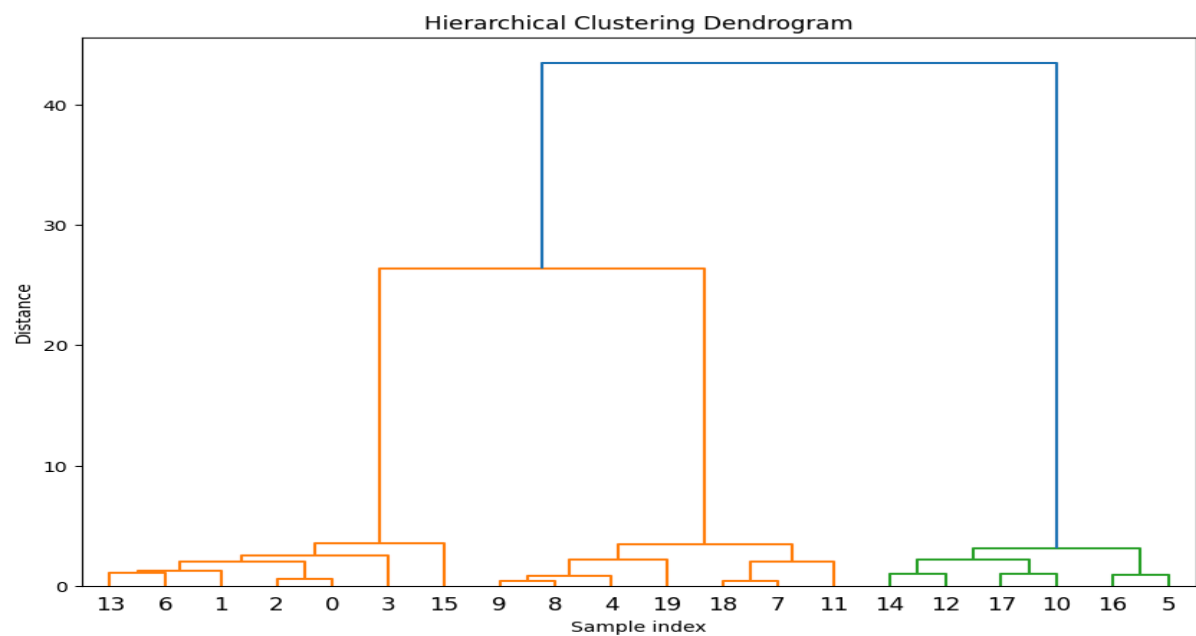
# Plot dendrogram
plt.figure(figsize=(10, 7))
dendrogram(linked,
            orientation='top',
            distance_sort='descending',
            show_leaf_counts=True)
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Sample index')
plt.ylabel('Distance')
plt.show()

# Assign cluster labels
cluster_labels = fcluster(linked, t=3, criterion='maxclust')

# Print cluster labels
print("Cluster Labels:")
print(cluster_labels)

# Plot the clusters with colors
plt.figure(figsize=(8, 5))
plt.scatter(X[:, 0], X[:, 1], c=cluster_labels, cmap='rainbow', s=100)
plt.title('Data points grouped into clusters')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.show()
```

Output-



Cluster Labels:

```
[3 3 3 3 2 1 3 2 2 2 1 2 1 3 1 3 1 1 2 2]
```

